Pooling, Normalisation & Activation Functions

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1 Pooling

Pooling is a type of operation added to Convolutional Neural Networks (CNNs) following individual convolutional layers. It is used to reduce the dimensionality of images by reducing the number of pixels in the output from the previous convolutional layer while retaining the most important information from the previous layer. It is a deterministic downsampling operation, meaning it has no learnable parameters (no weights or biases to update via gradient descent) but is rather defined by several hyperparameters.

1.1 Hyperparameters

Pooling contains the following key hyperparameters:

1. **Pooling type**: Defines how the pooling aggregation is computed. The most common pooling types are max and average pooling.
2. **Filter Size (f)**: Defines the dimensions of the window (usually a 2 x 2 or 3 x 3 square) that slides across a feature map to reduce its spatial size.
3. **Stride (s)**: Determines how many pixels the window shifts at each step.
4. **Padding (p)**: Not frequently used but adds border pixels (usually zero) to the input to control the spatial output size.

1.2 Mathematics

For a feature map with dimensions $n_h \times n_w \times n_c$, the dimensions of the output after a pooling layer are:

$$\left\lfloor \frac{n_h - f}{s} \right\rfloor + 1 \times \left\lfloor \frac{n_w - f}{s} \right\rfloor + 1 \times n_c$$

Where:

- n_h : input height

- n_w : input width
- n_c : number of channels
- f : filter size
- s : stride length

Note: *Number of channels remains unchanged.*

1.3 Different Pooling Types

There are others but the main three types of pooling are:

1. **Max Pooling**: Selects the maximum element from the region of the feature map covered by the filter. The output after max pooling is performed would be a feature map preserving the most prominent features obtained from the previous layer. This provides very good performance in most cases.

2. **Average Pooling**: Instead of selecting the maximum element from each region, this pooling types averages all elements present in the region. This gives us the average of the features present in a particular pooling dimension. This provides a more generalised representation of the input so it is useful where preserving the overall context is important.

3. **Global Pooling**: Reduces each channel in the feature map to a single value, producing a $1 \times 1 \times n_c$ output. This is equivalent to applying a filter size of $n_h \times n_w$. Two types of global pooling which are global max pooling and global average pooling. Global max pooling takes the maximum across the entire feature map. Global average pooling averages all values in the feature map. Global pooling is often used at the end of a CNN to reduce feature maps to a single value per channel, effectively summarising spatial information.

1.4 Examples

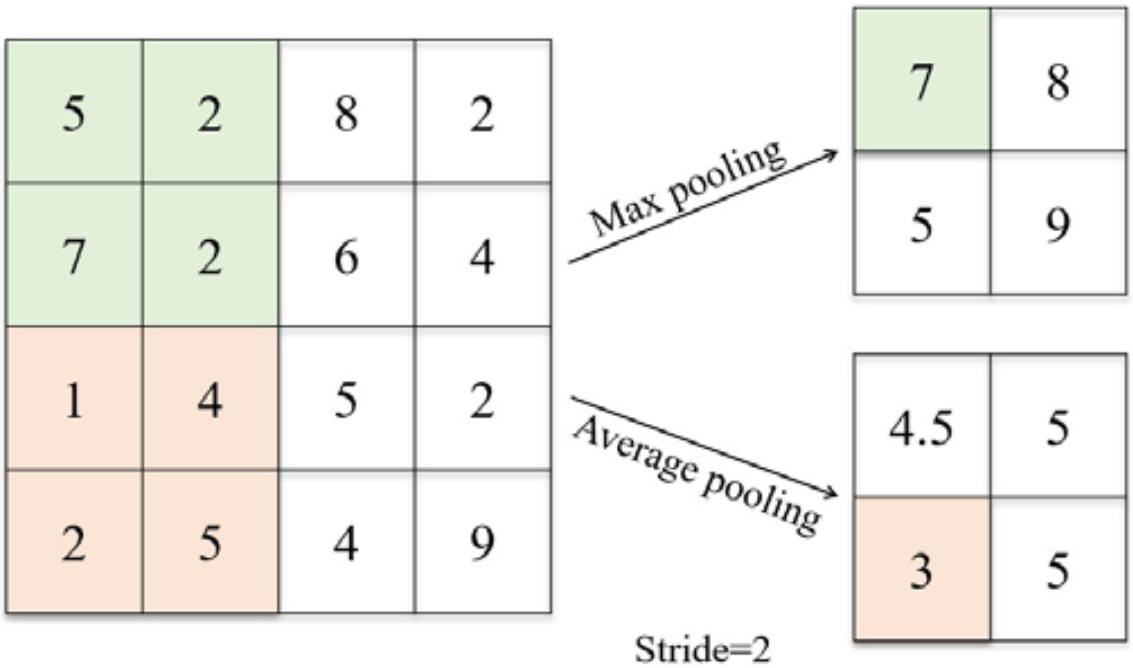

Figure 1: Example of max and average pooling using a 2×2 filter and a stride of 2.

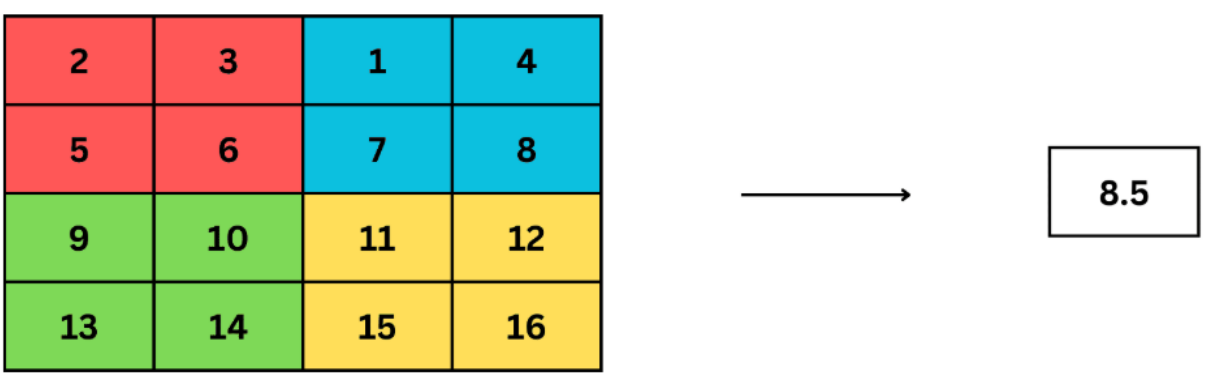

Figure 2: Example of global average pooling.

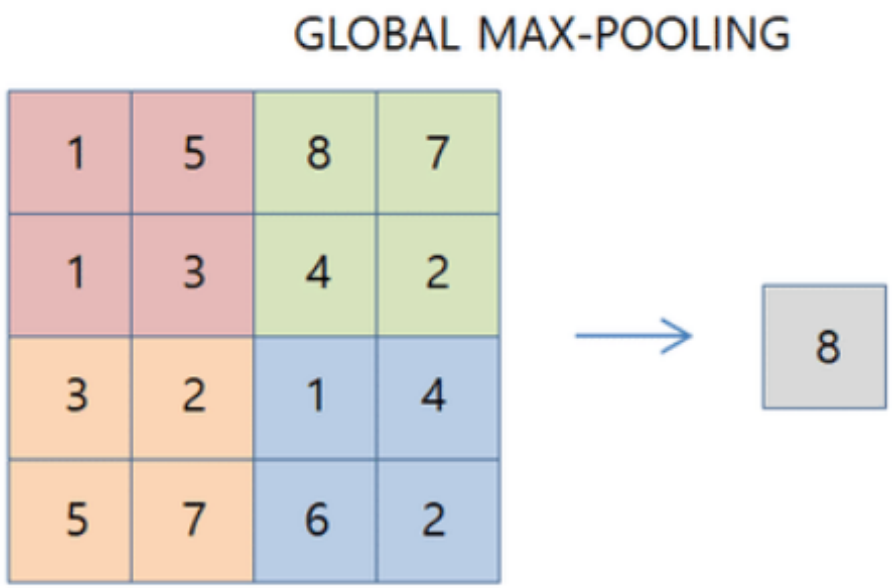

Figure 3: Example of global max pooling.

1.5 Uses & Limitations

Main advantages and uses of pooling are:

1. **Dimensionality Reduction & Efficiency**: Significantly reduces the spatial size of feature maps, which accelerates training as we lower the number of parameters and computations required.

2. **Translation Invariance**: Model is robust to small shifts, distortions, or translations in the input image.

3. **Controls Overfitting**: By summarising features and reducing the data complexity, it acts as a form of regularisation, helping to prevent the model from overfitting to training data.

4. **Feature Selection**: Retains important features (max pooling) or captures overall context (average pooling).

Main limitations of pooling are:

1. **Information Loss**: Since pooling is a form of data compression, pooling discards certain information, some of which may be important.

2. **Hyperparameter sensitivity**: Choice of hyperparameters is important to avoid excessive loss of information.

3. **Over-smoothing**: Improper design or excessive pooling can blur crucial features.

Examples of real-world applications that use pooling:

1. **Image Classification (e.g. VGG, AlexNet)**: Pooling layers are stacked after convolutional layers throughout the network. These are used to progressively shrink the spatial dimensions so that when data reaches the fully connected dense layers, the number of parameters is manageable without sacrificing the learned features.

2. **Face Recognition**: Systems like FaceNet use global average pooling before the embedding layer to produce a compact, fixed-size descriptor of a face regardless of input resolution. This makes the system robust to minor shifts or crops in the input image.

1.6 Quick Reference

1.6.1 Key Equations

Description	**Formula**
Output Height	$\left\lfloor \frac{n_h - f}{s} \right\rfloor + 1$
Output Width	$\left\lfloor \frac{n_w - f}{s} \right\rfloor + 1$
Channels (unchanged)	n_c
Global Pooling Filter Size	$f = n_h \times n_w$

1.6.2 Hyperparameter Typical Ranges

Hyperparameter	**Typical Values**
Filter size (f)	2×2 (most common), 3×3
Stride (s)	2 (most common), 1
Padding (p)	0 (rarely used in pooling)

1.6.3 Pooling Types at a Glance

Type	**Operation**	**Best Used When**
Max Pooling	Takes maximum in window	Preserving prominent features
Average Pooling	Averages all values in window	Preserving overall context
Global Max Pooling	Max across entire feature map	Feature presence detection
Global Average Pooling	Mean across entire feature map	End-of-network summarisation

1.6.4 Glossary

Translation Invariance The model produces the same or very similar output even if the input is slightly shifted. Pooling contributes to this but does not make a network fully invariant on its own.

Downsampling Reducing the spatial dimensions of a feature map. Pooling is one way to achieve this.

Feature Map The output of a convolutional layer which is a 3D volume of shape $n_h \times n_w \times n_c$ representing learned spatial features.

1.6.5 Common Misconceptions

- **Pooling has learnable parameters.** It does not. Pooling is entirely defined by its hyperparameters and no weights are updated during back propagation.
- **Pooling always improves performance.** Excessive or poorly configured pooling discards too much spatial information, which can harm tasks requiring precise localisation such as semantic segmentation.
- **Max pooling is always better than average pooling.** Max pooling tends to outperform when detecting whether a feature exists matters most. Average pooling can outperform when the overall distribution of activations carries meaning, particularly in global summarisation contexts.

1.7 Key Papers

1. *Evaluation of Pooling Operations in Convolutional Architectures for Object Recognition* [1].

   URL: https://link.springer.com/chapter/10.1007/978-3-642-15825-4_10

   Summary: This paper provides one of the first direct empirical comparisons of pooling operations within a fixed CNN architecture, evaluated on the Caltech-101 and NORB datasets. The authors compare max pooling against subsampling (similar to average pooling but is followed by multiplying by a trainable parameter and adding a trainable bias), finding that max pooling significantly outperforms subsampling in both convergence speed and recognition accuracy. They also investigate overlapping pooling windows and various window functions, concluding that neither offers consistent improvement over standard non-overlapping max pooling.

   Relevance: This paper is directly relevant as it provides controlled experimental evidence for why max pooling (and pooling in general) became the dominant choice in CNN architectures.

2. *Network In Network* [2].

   URL: https://arxiv.org/abs/1312.4400

   Summary: This paper introduces the Network In Network (NIN) architecture, which replaces standard linear convolutional filters with small multilayer perceptrons to better model non-linear relationships within local receptive fields. Its most significant contribution to pooling is the proposal of global average pooling (GAP) as a direct replacement for fully connected

layers at the end of a CNN. GAP reduces each feature map to a single value per channel, acting as a structural regulariser that enforces correspondence between feature maps and output categories, while substantially reducing parameters and overfitting.

Relevance: This paper is the original source proposing global average pooling, making it a foundational reference for this topic. Its influence is far-reaching, as GAP was subsequently adopted by major architectures including GoogLeNet and MobileNet, cementing it as a standard design choice in modern CNNs.

2 Normalisation

3 Activation Functions

References

[1] D. Scherer, A. Müller, and S. Behnke, "Evaluation of pooling operations in convolutional architectures for object recognition," in *Artificial Neural Networks – ICANN 2010* (K. Diamantaras, W. Duch, and L. S. Iliadis, eds.), (Berlin, Heidelberg), pp. 92–101, Springer Berlin Heidelberg, 2010.

[2] M. Lin, Q. Chen, and S. Yan, "Network in network," 2014.